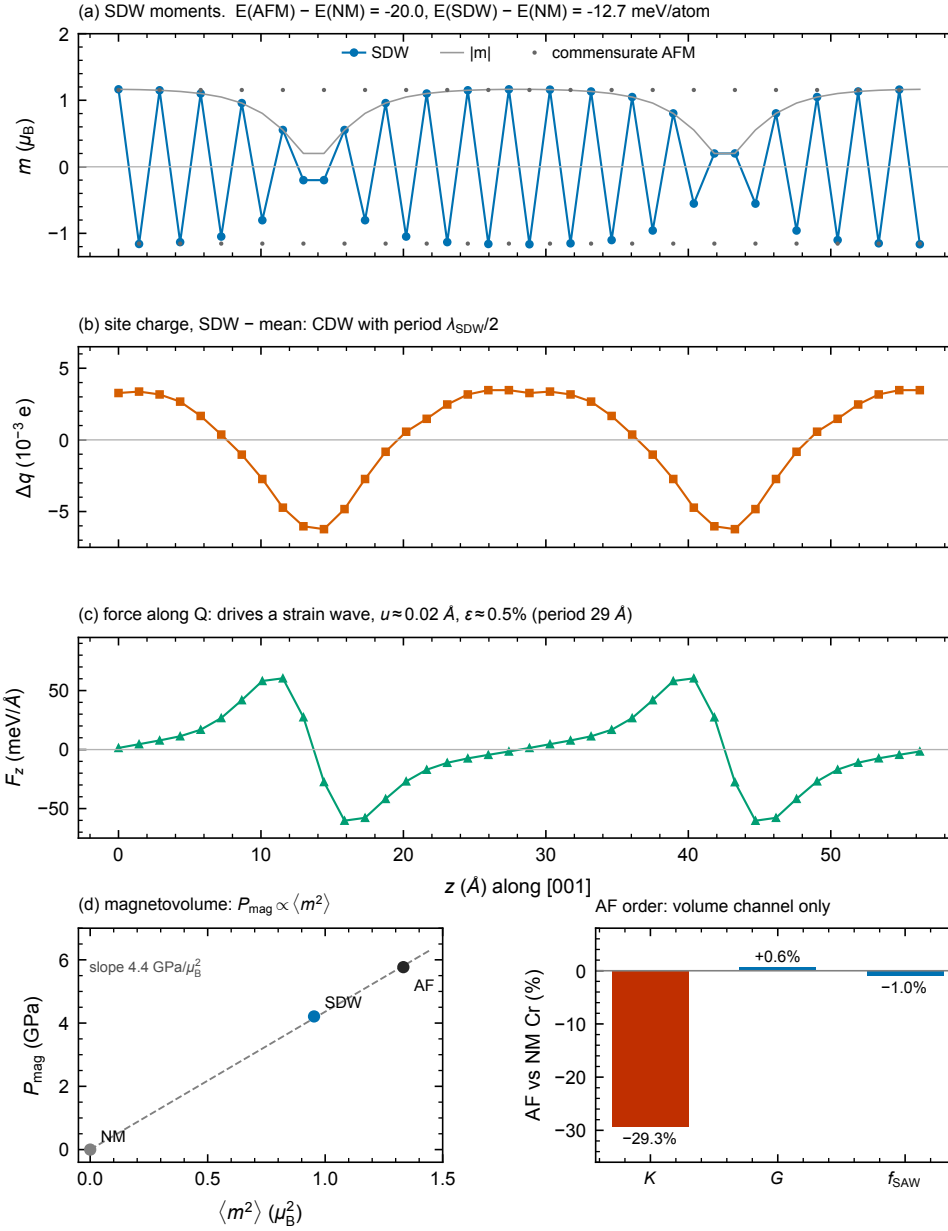
(a) SDW moments. E(AFM) − E(NM) = -20.0, E(SDW) − E(NM) = -12.7 meV/atom
SDW
|m|
commensurate AFM
m (μB)
(b) site charge, SDW − mean: CDW with period λSDW/2
Δq (10−3 e)
(c) force along Q: drives a strain wave, u ≈ 0.02 Å, ε ≈ 0.5% (period 29 Å)
Fz (meV/Å)
z (Å) along [001]
(d) magnetovolume: Pmag ∝ ⟨m2⟩
slope 4.4 GPa/μB2
NM
SDW
AF
Pmag (GPa)
⟨m2⟩ (μB2)
AF order: volume channel only
+0.6%
−1.0%
−29.3%
AF vs NM Cr (%)
K
G
fSAW